

Summer Term 2026 – Julián Méndez & Ricardo Langner

Interactive Information Visualization

Exercise: **Assignment 5**

Assignment 5

Goals

We will focus on **Large Display Visualization** and **Immersive Visualization**

- Degrees of freedom for the design of visualization applications
 - > *Analyze, ideate, design, create, sketch, ...*
 - > *Discuss, reflect, exchange, ...*
- General capabilities and possibilities
- Specific design options, choices, decisions
- Understand implications, consequences, and factors of influences

Assignment 5

Background: Imagine the following ...

- A **museum** asked you to design two new interactive installations/artifacts for an exhibition about the role and impact of data visualization in our daily lives.
 - *For this assignment: data visualization = visual representation of information and data, this includes information visualization and scientific visualization*
- Each installation in this exhibition aims to inform visitors about a specific aspect, characteristic, fact, or opinion of data visualizations – by showing/using data visualizations itself.
- Which information or data is displayed is up to you.
- Each of the two installations will be in its own room.
 - In the **first room**, there will be a **large interactive display**, around 8m wide and 3m high.
 - In the **second room**, several wireless **head-mounted displays** will be provided.
 - The exact technologies involved are up to you - you can basically wish for something ;)

Assignment 5

I. Installations/Artifacts

Create a first draft of the installation for the wall display. It is only about a rough idea. This means that, for example, the following questions are relevant:

- What take-home message(s), data insight(s), or fact(s) is/are presented?
- Which encoding(s) or visualization(s) is/are used?
- How can visitors interact in the space, with the information, or with each other?
- **Create a brief textual description of your idea, your installation.**
- **Create sketches or use AI tools to generate pictures of your envisioned installation.**
- **Remember: You need to do this twice, once for room 1 and once for room 2**

Assignment 5

II. Reflection

Reflect on the different degrees of freedom that have a significant influence on the design of the visualization application for the installation.

- **List and briefly describe 2-4 general design aspects/considerations**, design decisions to be made, or design implications for each of the following three categories:
 - **Output:** This is the perspective of how to present content
 - **Input:** This is the perspective of how to engage and interact with content
 - **Multi-user:** This is the perspective of an installation not being visited by just one person, neither at one point in time and certainly not aggregated over time
- Here are some example aspects:
 - How much content should be displayed, and how large (information density)?
 - What input modalities are used? (touch, speech, gaze, body/hand gestures, ...)
 - Does each visitor have their own display area? Do visitors have to stand still or move?
 - What can visitors manipulate, are there any menus or configuration options?
- **Remember: You need to do this twice, once for each installation**

Assignment 5

III. Comparison

Compare your findings regarding the two installations (the two different device environments).

- Which design aspects are relevant for both environments, which are more specific to one environment?
- **List at least two differences and similarities.**

Assignment 5

Summary of your tasks

- In your team
 1. Design two interactive installations/artifacts for a museum, one for a large display and one for HMDs
 2. Reflect on output, input, and multi-user aspects of your installations
 3. Compare your installations
- Remember, we want to do another **workshop on this assignment**. However, an active and fruitful discussion in such a workshop is only possible if everyone is prepared and can contribute something.

Exercise Interactive Information Visualization

Thank you!

Any questions?

Contact: julian.mendez2@tu-dresden.de
ricardo.langner@tu-dresden.de